REPORT: DRIVER DROWSINESS DETECTION SYSTEM

SUBMITTED BY:

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INTRODUCTION:

The Driver Drowsiness Detection System is a safety-oriented system designed to reduce road accidents caused by driver fatigue. Drowsiness during driving, especially on long routes or at night, can significantly affect a driver's reaction time and attention, increasing the risk of accidents.

This system works by continuously monitoring the driver's eye movements using sensors or a camera module. The basic principle is that an alert driver blinks normally for a short duration, while a drowsy driver tends to keep their eyes closed for a longer period. The system tracks this eye closure time and compares it with a predefined threshold value.

If the system detects that the driver's eyes are closed for longer than the set threshold, it identifies this as a sign of drowsiness. In response, the LEDs will glow and the speed of the vehicle will slow down gradually.

The eye blink sensor has two parts: The transmitter and the Receiver

When the person is Awake the light emitted by the transmitter is absorbed by the human eye and hence giving 0 as an output, but when he/she is in drowsy condition the light emitted by the transmitter is reflected back to the receiver and hence giving the output as 1 and making the LEDs and Motor deriver active for their respective works

By providing real-time monitoring and immediate alerts, the system helps in preventing accidents and enhances overall driving safety.

Components Required

- IR Eye Blink Sensor
- Arduino Uno
- Breadboard
- LED Bulbs
- 18650 Power Supply
- L293d Motor Driver
- DC Motors
- Chassis
- Jumper Wires
- Wheels

1. Normal Driving Condition (Driver Awake)

How it Works:

In this condition, the driver is fully alert and their eyes are open most of the time. The eye blink sensor is properly aligned with the driver's eyes and continuously monitors their activity.

Eye State: When the driver is awake, the eyes blink naturally for a very short duration The sensor detects this normal blinking pattern.

Signal Detection: Since the eyes are open most of the time, the sensor outputs a consistent signal indicating an active (awake) state. There is no continuous interruption in the signal.

Microcontroller Response: The Arduino receives this signal and determines that the driver is not drowsy, as the eye closure time is below the threshold value.

System Output: Since no drowsiness is detected, the system does not trigger any action. The LED remains OFF and the motors continue to run normally, indicating safe driving conditions.

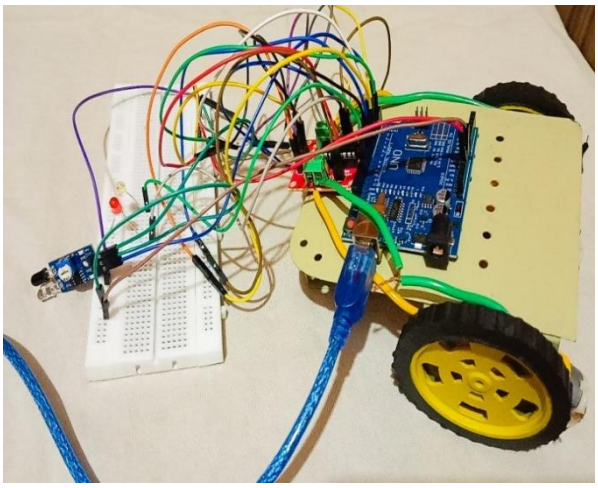


Figure 1(a): Circuit when the driver is awake

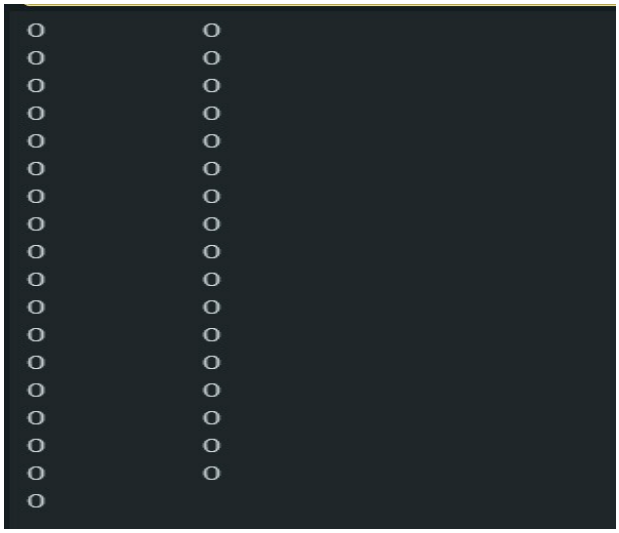


Figure 1(b): Output when the driver is awake

2. Drowsiness Condition (Eyes Closed)

How it Works:
In this condition, the driver starts feeling drowsy and their eyes remain closed for a longer duration. The eye blink sensor continuously monitors this change in eye activity.

Eye State: When the driver is drowsy, the eyes stay closed for a longer time compared to normal blinking. The sensor detects this prolonged eye closure.

Signal Detection: Due to continuous eye closure, the sensor output signal changes and indicates inactivity. This prolonged signal interruption is recognized as a sign of drowsiness.

Microcontroller Response: The Arduino processes this signal and measures the eye closure duration. If it exceeds the predefined threshold, the system identifies the driver as drowsy.

System Output: Once drowsiness is detected, the LED turns ON as an alert indication. At the same time, the L293D motor driver is controlled to stop the DC motors, bringing the vehicle to a halt and preventing potential accidents.

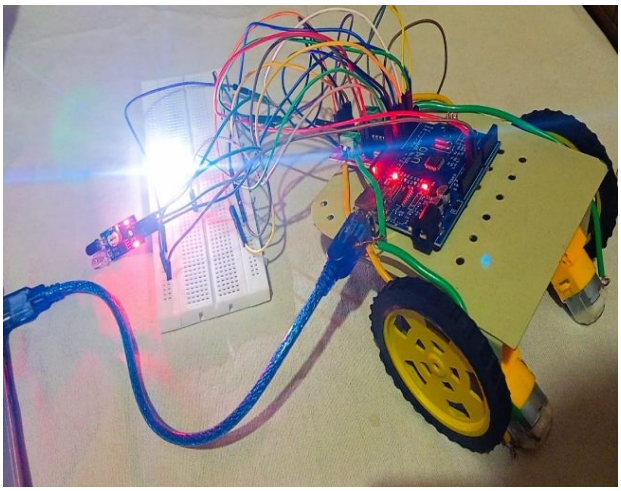


Figure 2(a): Circuit when the driver is sleepy

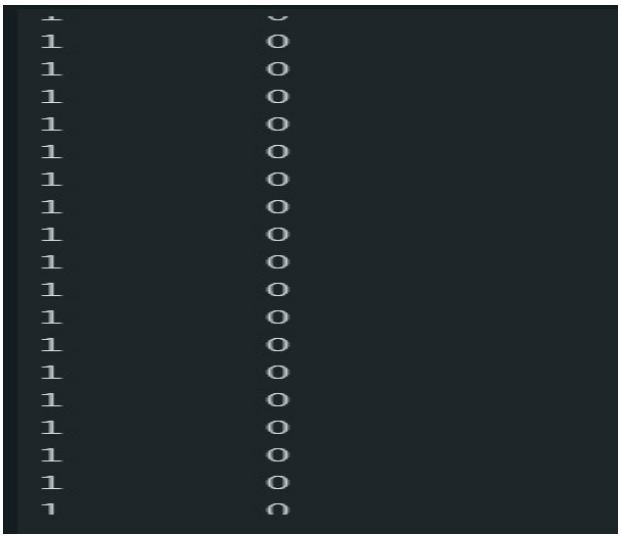


Figure 1(b): Output when the driver is sleepy

CONCLUSION:

The Driver Drowsiness Detection System is an effective and practical solution for improving road safety. By continuously monitoring the driver's eye activity, the system can detect fatigue and provide timely alerts. This project demonstrates how simple sensing and control mechanisms can be used to address real-world problems efficiently and enhance driver safety.

CODE FOR EXECUTION:

```
#include <Servo.h>
int sensor_pin = 2;
int Mot_En1 = 3;
int Mot_In1 = 4;
int Mot_In2 = 5;
int Mot_En2 = 6;
int Mot_In3 = 7;
int Mot_In4 = 8;
int buz_pin = 9;
int led_pin1 = 10;
int led_pin2 = 11;
static int cntr;
static bool deb_out, sleep_detect;
void setup()
{
  Serial.begin(9600)
  pinMode(sensor_pin, INPUT);
  pinMode(Mot_In1, OUTPUT);
  pinMode(Mot_In2, OUTPUT);
  pinMode(Mot_En1, OUTPUT);
  pinMode(Mot_In3, OUTPUT);
  pinMode(Mot_In4, OUTPUT);
  pinMode(Mot_En2, OUTPUT);
  pinMode(led_pin1, OUTPUT);
  pinMode(led_pin2, OUTPUT);
  pinMode(buz_pin, OUTPUT);
  digitalWrite(Mot_In1, LOW);
  digitalWrite(Mot_In2, HIGH);
  analogWrite(Mot_En1, 255);
  digitalWrite(Mot_In3, LOW);
  digitalWrite(Mot_In4, HIGH);
  analogWrite(Mot_En2, 255);
}
void loop()
{
  int sensor_data = digitalRead(sensor_pin);
  Serial.print(sensor_data);
```

```
  Serial.print(" \t");
  sleep_detect = deb_cntr(sensor_data, 1200);
  if(sleep_detect == true)
  {
    Motor_Ctrl();
    while(1)
    {
      digitalWrite(Mot_In1, LOW);
      digitalWrite(Mot_In2, LOW);
      digitalWrite(Mot_In3, LOW);
      digitalWrite(Mot_In4, LOW);
      Led_Buzz_Ctrl();
    }
  }
  Serial.println(sleep_detect);
}
bool deb_cntr(bool inp, int count)
{
  if(inp == false)
  {
    cntr+=1;
    if(cntr >= count)
    {
      deb_out = true;
    }
  }
  else
  {
    deb_out = false;
    cntr = 0;
  }
  return deb_out;
}
void Led_Buzz_Ctrl()
{
  digitalWrite(led_pin1, HIGH);
  digitalWrite(led_pin2, HIGH);
  digitalWrite(buz_pin, HIGH);
  delay(250);
  digitalWrite(led_pin1, LOW);
  digitalWrite(led_pin2, LOW);
  digitalWrite(buz_pin, LOW);
  delay(250);
}
```

```
void Motor_Ctrl()
{
  for(int i=255; i>=0; i--)
  {
    analogWrite(Mot_En1, i);
    analogWrite(Mot_En2, i);
    delay(5);
```